US ACCIDENTS

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Big data course project

If accident severity could be identified in advance, Emergency Medical Services, Departments of Transportation, and Navigation Services could improve emergency response, traffic control, and route planning to reduce casualties and congestion.

Historical accident data together with environmental and road conditions could help evaluate the likely severity of new traffic incidents.

Dataset

KAGGLE

US ACCIDENTS (2016–2023)

Nationwide US table of logged traffic incidents with time, place, weather, road context, and a four-step severity of traffic disruption.

7,7M rows

49 states

46 columns

Timestamps

accident interval in the local timezone.

Geospatial data

GPS start/end (end pair optional)

Target: severity (integer 1–4)

impact on traffic flow:
1 short delay → 4 long closure

DATA PIPELINE

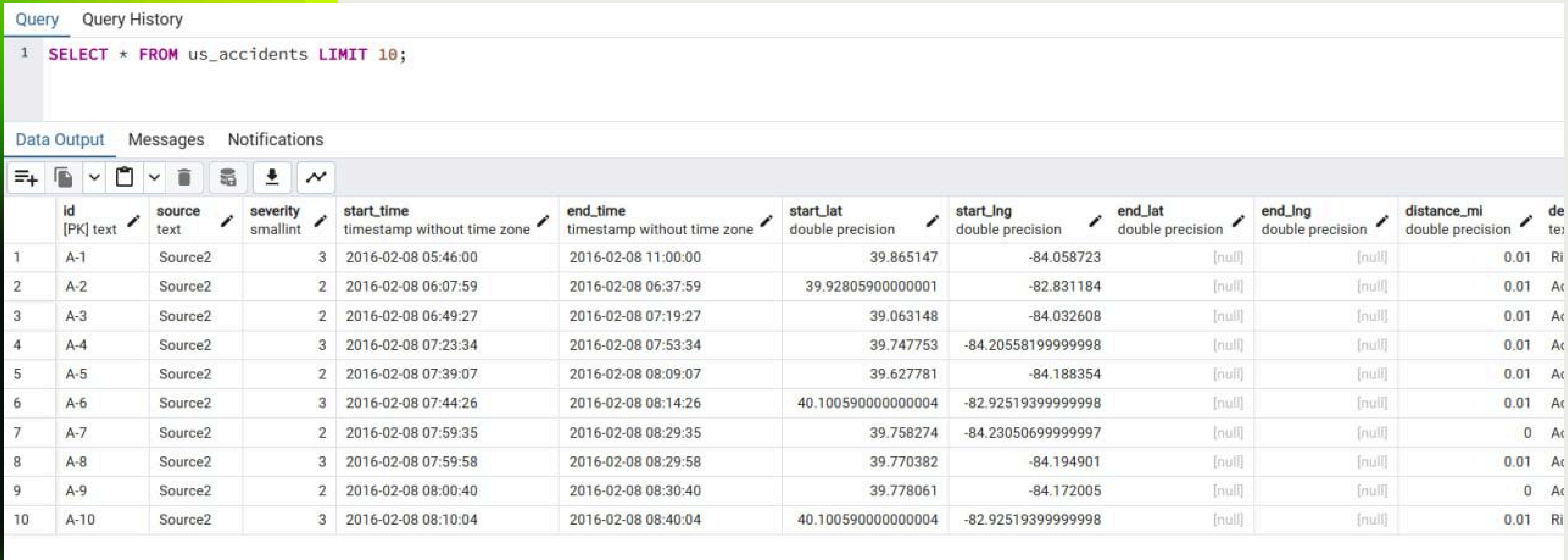
DATA COLLECTION/INGESTION	01.
DATA STORAGE USING HIVE AND AVRO	02.
EXPLORATIVE DATA ANALYSIS THROUGH HIVEQL QUERIES	03.
PREDICTIVE DATA ANALYSIS USING MACHINE LEARNING	04.

Data ingestion

- Ingested the raw flat export from Kaggle.
- Cleaned data in Python in 200k-row chunks: deduplication, missing-value handling, and coordinate validation.
- Bulk-loaded staging data into PostgreSQL.
- Exported PostgreSQL to HDFS via Sqoop as Avro with Snappy compression and embedded Avro schema.

Schema-safe binary export & splittable compression

Avro preserves types on ingest; Snappy stays splittable for the cluster and cuts disk use.



Query | Query History

```sql
SELECT * FROM us_accidents LIMIT 10;
```

Data Output | Messages | Notifications

	id [PK] text	source text	severity smallint	start_time timestamp without time zone	end_time timestamp without time zone	start_lat double precision	start_lng double precision	end_lat double precision	end_lng double precision	distance_mi double precision	de te
1	A-1	Source2	3	2016-02-08 05:46:00	2016-02-08 11:00:00	39.865147	-84.058723	[null]	[null]	0.01	Ri
2	A-2	Source2	2	2016-02-08 06:07:59	2016-02-08 06:37:59	39.92805900000001	-82.831184	[null]	[null]	0.01	Ac
3	A-3	Source2	2	2016-02-08 06:49:27	2016-02-08 07:19:27	39.063148	-84.032608	[null]	[null]	0.01	Ac
4	A-4	Source2	3	2016-02-08 07:23:34	2016-02-08 07:53:34	39.747753	-84.20558199999998	[null]	[null]	0.01	Ac
5	A-5	Source2	2	2016-02-08 07:39:07	2016-02-08 08:09:07	39.627781	-84.188354	[null]	[null]	0.01	Ac
6	A-6	Source2	3	2016-02-08 07:44:26	2016-02-08 08:14:26	40.100590000000004	-82.92519399999998	[null]	[null]	0.01	Ac
7	A-7	Source2	2	2016-02-08 07:59:35	2016-02-08 08:29:35	39.758274	-84.23050699999997	[null]	[null]	0	Ac
8	A-8	Source2	3	2016-02-08 07:59:58	2016-02-08 08:29:58	39.770382	-84.194901	[null]	[null]	0.01	Ac
9	A-9	Source2	2	2016-02-08 08:00:40	2016-02-08 08:30:40	39.778061	-84.172005	[null]	[null]	0	Ac
10	A-10	Source2	3	2016-02-08 08:10:04	2016-02-08 08:40:04	40.100590000000004	-82.92519399999998	[null]	[null]	0.01	Ri

Table optimal for analytics

Single staging table us_accidents: accident metadata, geography, weather, and road-feature flags

Data storage

- Created a Hive warehouse layer and an external Avro table on top of HDFS.
- Created an optimized table partitioned by state, bucketed by severity into 4 buckets and stored as ORC.
- Ran 7 HiveQL exploratory queries and exported EDA aggregates for charts.

DEMO

CHECK THE RESULTS

EDA findings

01. BIMODAL DISTRIBUTION

Accidents follow a bimodal time distribution with the peaks at 7-8am and 3-6pm (accidents depend on commute road clogging)

02. CLASS IMBALANCE

Severity 2 takes up the majority of the dataset - ML model should account for this

03. WEATHER CONDITION IMPACT

Extreme weather conditions do not increase accidents as people stay at home instead; moderate weather tends to be more dangerous

04. ROAD FEATURE IMPACT

Intersections and merging zones tend to be the most dangerous, most common, or both

Feature pipeline

Learning problem

Multiclass severity prediction using the 4 traffic impact levels. Map the original severity numbers to class labels for Spark ML

Analysis details

Run in Apache Spark on YARN cluster

Time encoding

Uses sine and cosine

Location encoding

Converted from lat and long to 3-D Earth coordinates

Categorical encoding

Categorical fields combined into numeric indices; columns combined into one feature vector

Models

Model	numClasses	numFeatures	Accuracy	F1-score
Logistic Regression	4	41	79.95%	76.19%
Random Forest	**4**	**41**	**81.86%**	**76.72%**

HYPERPARAMETER TUNING

CrossValidator and 3 folds
Each fold fit on training split only

SELECTION SCORE

MulticlassClassificationEvaluator
with F1 score and accuracy

2 MULTICLASS MODELS

MULTINOMIAL LOGISTIC REGRESSION

Reg. Param: {0.001, 0.01, 0.05}
elasticNet: {0.0, 0.5}

RANDOM FOREST

numTrees: {40, 80, 120}
maxDepth: {6, 10}

60%
TRAIN

40%
TEST

DEMO

CHECK THE RESULTS

Conclusions

We learned how to

- Build a scalable end-to-end pipeline for 7.7M records;
- Optimize Hive storage using ORC, partitioning, and bucketing;
- Implement advanced Spark MLlib feature engineering;
- Transform complex model outputs into visual storytelling.

Challenges

- Handling extreme class imbalance (79% Severity 2);
- Working in cluster in the first time;
- Cluster ingested timezones incorrectly the first time; had to specify timezone in Hive
- Superset is not intuitively understandable.

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